

2(a)(i)	<u>directional</u> ;	1
2(a)(ii)	<i>any three from</i> 1 lactose / milk (products), acts as a selection pressure or <i>ref. to</i> reliance on milk (products) ; 2 (selective) <u>advantage</u> to digest, lactose / milk (products) or individuals with lactose persistence have a (selective) <u>advantage</u> ; 3 (those individuals) more likely to, survive / reproduce ; or a 4 passed on the (mutated) allele (to their offspring) ; 5 over, time / many generations, the allele frequency increased ;	3
2(b)(i)	$q^2 = 58 \div 166$ or 0.349 ; A 0.35 $q = \sqrt{0.349}$ or 0.591 ; A 0.59 $p = 0.409$ or 0.41 ;	3
2(b)(ii)	<i>any two from</i> 1 test population is (too) small ; 2 test population, not representative of general population / not random / is biased ; 3 migration / ethnic origin ;	2
2(c)(i)	<i>any two from</i> 1 proteins that bind to DNA ; 2 binds to the promoter ; A enhancers 3 control, gene expression / transcription / mRNA synthesis ; 4 allow attachment of RNA polymerase to DNA ;	2
2(c)(ii)	promoter / enhancer / silencer / insulator ;	1
2(c)(iii)	1 most genes are regulatory genes (in the genome) ; or a 2 mutations in regulatory genes less likely to be, harmful / selected against / affect survival ; or a	2